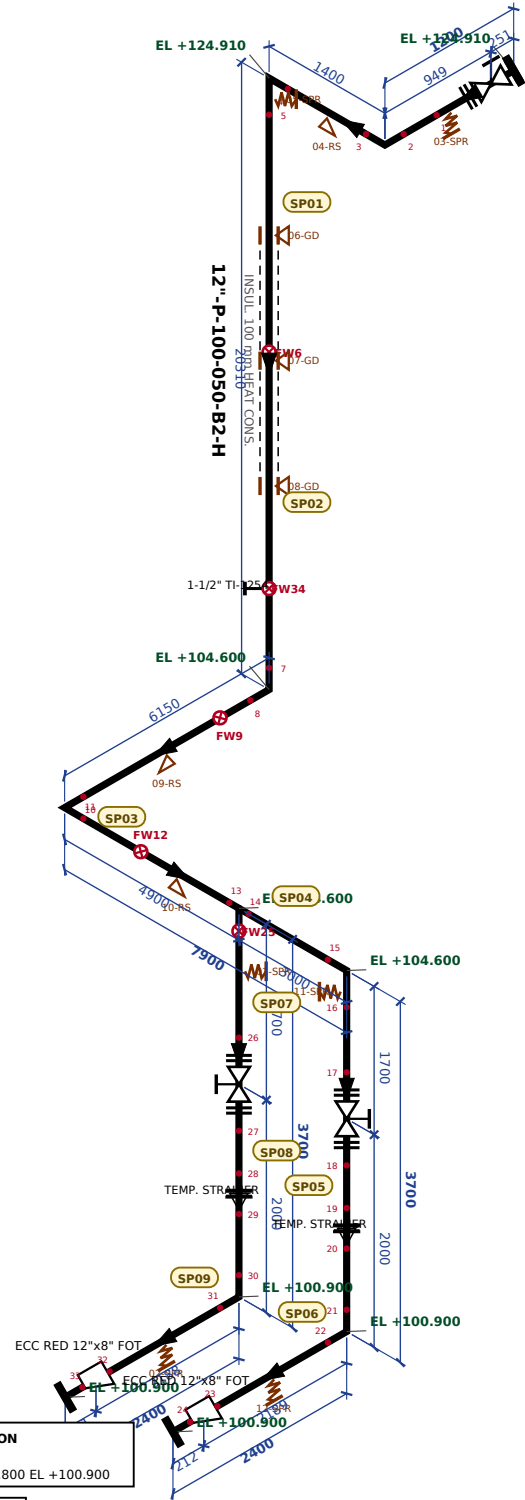
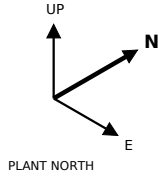


ISOMETRIC VIEW - NOT TO SCALE - DIMENSIONS IN mm, ELEVATIONS / COORDINATES IN m (PLANT GRID)



C-101 BPA DRAW (TRAY 25)
12" NOZZLE
E 95.000 N 94.550 EL +124.910

P-108B SUCTION
8" NOZZLE
E 101.500 N 84.800 EL +100.900

P-108A SUCTION
8" NOZZLE
E 98.500 N 84.800 EL +100.900

LINE NUMBER 12"-P-100-050-B2-H	FROM C-101 tray 25	TO P-108A/B	PIPING CLASS B2 - 300# 5Cr-1/2Mo ASTM A335 P5
DESIGN P / T 3.5 barg / 345 C	OPER. P / T 1.69 barg / 313.9 C	TEST P (HYDRO) 6.4 barg (1.5xPxST/S=1.21)	PIPE WALL STD (9.53 mm)
INSULATION H 100 mm	PWHT YES - all thk (P-No.5B, B31.3 Tbl)	NDE 100 % RT of BW; 100 % MT/PT of fillets;	STRESS CAT 1 - T > 300 C & NPS >= 6; T > 200 C & NPS >= 1
P&ID CFU-100-PR-PID-007	FLUID / PHASE P / L BPA draw to pumps	PAINT HT silicone/Al (insul.)	SPOOLS / WELDS 9 spools; 29 shop / 5 field

BILL OF MATERIALS (THIS SHEET)

PT	DESCRIPTION	NPS	QTY
1	FLANGE WN RF CL300 B16.5, bore STD, ASTM A182 F5	12"	9
2	PIPE STD, BE, ASTM A335 P5 SMLS	12"	38.7 m
3	ELBOW 90 LR, BW, STD, ASTM A234 WP5	12"	7
4	TEE EQUAL, BW, STD, ASTM A234 WP5	12"	1
5	REDUCER ECC, BW, STD, ASTM A234 WP5	12" x 8"	2
6	FLANGE WN RF CL300 B16.5, bore SCH 40, ASTM A182 F5	8"	2
7	SOCKOLET 3000# (branch), ASTM A182 F5	1-1/2"	1
8	ERECTION: GATE VALVE OS&Y CL300 RF, A217 C5, trim 5	12"	3
9	TEMPORARY CONE STRAINER (commissioning)	12"	2
10	GASKET SPW 321/graphite, inner+outer ring, CL300 B16.20	12"	10
11	STUD BOLT SET (nuts) A193 B7 / A194 4, CL300	12"	10
12	PIPE SUPPORTS (std. dwgs): guide x3, rest x3, spring x6	-	12

CUT LENGTHS (mm, ex. weld gap) / SPOOLS

PIECE	NPS	L	PIECE	NPS	L	PIECE	NPS	L
SP01-1	12"	112	SP01-2	12"	486	SP01-3	12"	9350
SP02-1	12"	10046	SP02-2	12"	950	SP03-1	12"	4286
SP03-2	12"	2150	SP04-1	12"	2039	SP04-2	12"	2289
SP04-3	12"	863	SP05-1	12"	191	SP06-1	12"	714
SP06-2	12"	1630	SP07-1	12"	1066	SP08-1	12"	191
SP09-1	12"	714	SP09-2	12"	1630			

SPOOL DEVELOPED LENGTHS (transport limit 12.0 x 3.0 x 3.0 m):

SP01=11.9m SP02=11.9m SP03=7.3m SP04=6.7m SP05=0.4m SP06=3.7m SP07=1.2m SP08=0.4m SP09=3.7m

NOTES

- Pipe/fittings/flanges per class B2 (CFU-000-PI-SPC-001); BW fittings ASME B16.9, flanges B16.5/B16.47A RF.
- Hydrotest 6.4 barg per ASME B31.3 345.4 (vents at high points, drains at low points).
- PWHT: YES - all thk (P-No.5B, B31.3 Tbl 331.1.1) 705-770 C. NDE: 100 % RT of BW; 100 % MT/PT of fillets; HB <= 241; 100 % PMI.
- Supports: RS rest/shoe, GD guide, ANC anchor, SPR variable spring - tags PS-<area><seq>-nn.
- Welds: dot = shop weld, crossed circle FW = field weld; spools <= 12 m for transport.
- Pump suction: no pockets, header falls to pumps; eccentric reducers 12"x8" FLAT ON TOP; temporary straine
- Pump nozzle 8" assumed (API 610 vendor data pending).
- Draw turns west to clear BPA return riser and C-101 OH line drop.
- Hot pump (345 C): spring support at first support off column; warm-up lines at pumps.

REV	DATE	DESCRIPTION
A	2026-10-02	Issued for review (FEED)

GENERIC REFINERY - GULF COAST
100 KBPSD CRUDE & VACUUM DISTILLATION UNIT

PIPING ISOMETRIC 12"-P-100-050-B2-H

C-101 BPA DRAW TO P-108A/B

DISCIPLINE	STAGE	SCALE	SHEET
PIPING	FEED	NTS	1 OF 1
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			
CFU-100-PI-ISO-007			
REV			
A			